Lesson Planning

Complete Lesson Plan Book

Class: Class 9

Subject: Chemistry

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Acids, Bases and Salts: Everyday Examples

Learning Objectives

• Students will define acids and bases and give two everyday examples of each.

• Students will explain how indicators like litmus show whether a solution is acidic or basic.

• Students will describe neutralisation with a real-life example such as an antacid tablet.

Key Concepts

• An acid is a substance that tastes sour and turns blue litmus red; lemon juice, vinegar and stomach acid are common acids.

• A base is a substance that feels slippery and turns red litmus blue; soap solution, baking soda solution and toothpaste are common bases.

• The pH scale runs from 0 to 14: below 7 is acidic, 7 is neutral and above 7 is basic.

• When an acid reacts with a base, they neutralise each other and form a salt and water.

• Common salt, sodium chloride, is the product of hydrochloric acid reacting with sodium hydroxide.

Teaching Methodology (Step by Step)

1. Ask students why lemon tastes sour and soap feels slippery, and collect their answers on the board.

2. Define acid and base and write two everyday examples of each, then test them with litmus paper in a demonstration.

3. Explain the pH scale with a number line from 0 to 14 and place lemon juice, pure water and soap solution on it.

4. Show neutralisation on the board: acid plus base gives salt plus water, with the hydrochloric acid and sodium hydroxide example.

5. Connect to daily life: an antacid tablet neutralises extra stomach acid, which is why it relieves acidity.

Classroom Activities

• Litmus testing: students test lemon juice, vinegar, soap solution and water with litmus paper and record the colour change.

• pH sorting: cards naming common substances are placed on a 0 to 14 number line in the correct acidic, neutral or basic zone.

• Antacid story: students write in two lines how an antacid tablet works inside the stomach.

Assessment Questions

1. Vinegar turns blue litmus red. Is vinegar an acid or a base? Give one more example of the same kind.

2. A solution has pH 9. Is it acidic, neutral or basic? Name one household substance with a similar pH.

3. Write the word equation for the reaction between hydrochloric acid and sodium hydroxide.

Homework

• List three acids and three bases found in your kitchen and note how you identified each one.

• Explain in your own words why toothpaste helps protect teeth from acid attack.

Time Allocation

40 minutes

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